CSE400 Fundamentals of Probability in Computing

Domain : Finance
Group: s1_g1_fin

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Project Context and Motivation

- **Problem:** Portfolio Optimization under uncertainty
- Asset returns are modeled as random variables
- **Objective:**
  a. Maximize expected return
  b. Minimize risk

In M3: We used deterministic approach using optimization functions under given constraints

Motivation:

- Introduce randomized algorithm (PSO) to improve decision-making, can also provide with multiple portfolios which has similar returns

Deterministic Baseline Overview

- Based on **mean-variance framework optimization**
- **Inputs:**
  - Expected return vector (μ)
  - Covariance matrix (Σ)
- **Objective:**
  - Maximize sharpe ratio
- **Constraints:**
  - $w_i >= 0$
  - $\Sigma w_i = 1$
- **Limitations:**
  - Does not account for sudden shocks in market
  - Does no implement uncertainty and variability
  - Assumes returns and covariance does not change over time

System Pipeline Overview

- **System Pipeline:**
  - Get historical data
  - Compute expected returns and covariance
  - Generate a equally weighted portfolio
  - Optimize the portfolio using an optimization function (scipy.optimize.minimize) based on a given parameter
  - Evaluate the risk and returns, and the sharpe ratio.

- **Assumptions:**
  - Returns follow a normal distribution
  - Historical data approximated the future behaviour

Sources of Uncertainty

- **Market Variability:**
  - Price fluctuations caused by supply and demand
  - Economic and geopolitical influences
  - Volatility in financial markets
- **Data Uncertainty:**
  - Limited historical data
  - Measurement noise in market prices
  - Estimation errors in statistical parameters
- **Dependency Uncertainty:**
  - Asset returns are correlated
  - Correlations may change over time
  - Sector-wide events influence multiple assets simultaneously

Random Variables in the System

- **Key Random Variables**:
  - Asset Returns (R_i): modeled as normal random variables
  - Portfolio Return (R_p): R_p = w` R
  - Portfolio Risk (σ□): Derived from covariance of the given assets
- **Assumptions:**
  - Returns follow a normal distribution
  - Historical data approximated the future behaviour

Randomized Algorithm Design

Objectives:

- Maximize Sharpe Ratio
  - Weights ≥ 0
  - Sum of weights = 1

PSO Representation:

- Particle → Portfolio weights
- Random selection from sample space of weights

Randomization (Key Features)

- Random initialization (`Dirichlet`)
- Random factors r1,r2
- Noise added to velocity
- Mutation (random restart)

Learning

- pBest → best per particle
- gBest → selected from top candidates
- Independent sample spaces

Probabilistic Reasoning Behind the Algorithm

Portfolio Computation

- Return = $w^T \mu$
- Risk = $\sqrt{w^T \Sigma w}$
- sharpe ratio= (return-risk_free_rate/risk)

  → risk_free_rate = 0.02

Uncertainty

- Market returns are stochastic
- Optimal weights are not fixed

Why Randomization Helps

- Explores multiple portfolios
- Noise + mutation avoid local optima
- Stochastic gBest improves diversity

Implementation of the Randomized Strategy

1. Generate random portfolios using Dirichlet sampling *(n_samples = 5000)*
2. Initialize PSO particles (40 particles)
3. Compute Sharpe ratio for each portfolio
4. Update velocity (inertia + cognitive + social + noise)
5. Update particle positions
6. Apply mutation (random restart with probability)
7. Normalize weights (sum = 1, non-negative)
8. Update pBest and gBest
9. Repeat for 5000 iterations and sample space

Evaluation Metrics

Expected Return:

- Measures profitability of portfolio
- np.(w, mu)

Risk (Standard Deviation):

- Measures variability in returns
- sqrt($w^T \Sigma w$)

Sharpe Ratio:

- Measures risk-adjusted return
- (return - risk_free_rate) / risk

Comparative Results

Output

- Optimized portfolio weights (gBest)
- Expected return, volatility, Sharpe ratio

Observation

- PSO finds better allocation than equal weights
- Higher Sharpe ratio achieved
- Optimized weights are **close to deterministic baseline but more efficient**
- Results may vary slightly across runs due to randomness

Key Insights

Effect of Randomization

- Random initialization (Dirichlet) ensures diverse portfolios
- Noise + mutation improve exploration of search space

Improved Optimization

- PSO iteratively refines portfolio using pBest and gBest
- Achieves better Sharpe ratio than fixed allocation

Trade-off

- Better performance comes with higher computation
- Results may vary slightly across runs